The appendix provides proofs and additional analysis in Appendices A and B, evaluation details in Appendix C, and sentence-level prefix experiments in Appendix D.

A Proofs and finite-budget analysis

A.1 Proof of Theorem 1

Define the self-consistency continuation value at any history by backward recursion.

$$V_j^S(h) = \int V_{j+1}^S(h,u)K_j^S(du \mid h), \qquad V_J^S(h) = \mathbf{1}\{\text{final vote in } h \text{ is correct}\}.$$

The notation (h, u) includes the deterministic bookkeeping after the next window. Under path-consistency, conditional expectation gives

$$\mathbb{E}_P[V_{j+1}^S(H_{j+1}) - V_j^S(H_j) \mid H_j = h] = d_j(h).$$

Taking expectations and summing over j telescopes. The initial value is A_S and the expected terminal value is A_P, proving Equation 4. This argument needs only bounded measurable values and a finite horizon. It does not require independent windows, a binary answer distribution, or a correct selected prefix. Since $0 \le V_{j+1}^S \le 1$, the standard variational definition

$$\mathrm{TV}(P,Q) = \sup_{0\le f\le 1} |\mathbb{E}_P f - \mathbb{E}_Q f|$$

implies $|d_j(h)| \le \mathrm{TV}(K_j^P, K_j^S)$. The triangle inequality and the fact that two probabilities differ by at most one prove Equation 5.

Scope for stopping controllers. The displayed result compares fixed-budget self-consistency and path-consistency. A stopping controller can be represented by an absorbing terminal state and padding to a common finite maximum horizon. Both the reference value and the reference transition kernel must then include that parent's stopping rule. This extension does not allow replacing a confidence-weighted parent's final vote by a majority vote while claiming to compare the same parent.

A.2 Proof of Theorem 2

Fix a competitor a. The differences $D_i = X_i(a) - \mu_i(a)$ are martingale differences. Conditional on $\mathcal{F}_{i-1}$, their values lie in an interval of width at most two, since $X_i(a) \in [-1, 1]$. The conditional bounded-variable exponential inequality therefore gives

$$\mathbb{E}[\exp(-\lambda D_i) \mid \mathcal{F}_{i-1}] \le \exp(\lambda^2/2), \quad \lambda > 0.$$

This inequality follows by the convex upper bound on the exponential between the two interval endpoints and maximizing over a zero-mean two-point distribution. Iterating conditional expectations yields $\mathbb{E}\exp(-\lambda\sum_i D_i) \le \exp(B\lambda^2/2)$. If the correct answer fails to outvote a, then $\sum_i X_i(a) \le 0$. By the almost-sure lower bound on accumulated conditional means, this implies $\sum_i D_i \le -\Gamma_a$. Markov's inequality now bounds its probability by

$$\exp(-\lambda\Gamma_a + B\lambda^2/2).$$

Taking $\lambda = \Gamma_a/B$ gives $\exp(-\Gamma_a^2/(2B))$. If the final vote is incorrect or tied, at least one incorrect category has count no smaller than the correct category. Union-bounding these events proves the result. Voting failure is covered by the union of competitor events.

What the assumption rules out. An unconditional gap $P(A_i = a^\star) > P(A_i = a)$ is insufficient. Conditioning on an early misleading prefix may reverse the gap, violating the almost-sure lower bound. Similarly, the observed mode need not equal $a^\star$. The Beta gate operates on the mode, whereas the theorem refers to the externally defined correct answer. An empirical evaluation of the assumption needs correctness information and conditional resampling beyond the current trajectories.

A.3 Exact binary illustration without Taylor expansion

For comparison with a single-promotion analysis, suppose n root answers are independent Bernoulli with correctness probability p_0, and the remaining $r = B - n$ answers are conditionally independent with correctness probability q_k after observing k correct root answers. Fix a tie-as-failure convention. Define

$$Q_r(p, t) = \sum_{j=\max(0,t)}^{r} \binom{r}{j} p^j (1-p)^{r-j},$$

with $Q_r(p, t) = 0$ for $t > r$ and $Q_r(p, t) = 1$ for $t \leq 0$. Then

$$A_{\mathrm{PC}} = \sum_{k=0}^{n} \binom{n}{k} p_0^k (1-p_0)^{n-k} Q_r(q_k, \lfloor B/2 \rfloor + 1 - k), \tag{7}$$

$$A_{\mathrm{SC}} = Q_B(p_0, \lfloor B/2 \rfloor + 1). \tag{8}$$

Both parts contribute to the final vote. A gate that does not promote sets $q_k = p_0$ for that history. The formula is exact for this idealized model, including its tie convention; it is not a model of all levels of path-consistency.

For a concrete failure construction take $B = 32, n = 4, p_0 = 0.6, C = 0.8$. If the four root answers are all wrong, their agreement score is $1 - 2^{-5} = 0.96875$. Let the selected wrong prefix force all subsequent answers to be wrong ($q_0 = 0$), while all other histories retain $q_k = 0.6$. The accuracy loss relative to self-consistency is exactly

$$(0.4)^4 Q_{28}(0.6, 17) > 0.$$

Numerically, self-consistency accuracy in this construction is 0.835203095 and the intervention accuracy is 0.821096895, a loss of 1.41 percentage points. Thus even a model whose unprefixed correctness probability exceeds one half can be harmed by a gate-passing prefix. The construction does not assume symmetric beneficial and harmful effects. That symmetry is an additional modeling assumption, not an inference-time property. More generally, a consensus-only controller has no access to the sign of a prefix's correctness effect.

A.4 Shared-prefix dependence

Let a random shared prefix Z induce correctness probability p_Z, and let Y_1, Y_2 be correctness indicators of two suffixes that are conditionally independent given Z. Then $\mathbb{E}[Y_1 Y_2 \mid Z] = p_Z^2$ and $\mathbb{E}Y_i = \mathbb{E}p_Z$, so

$$\mathrm{Cov}(Y_1, Y_2) = \mathrm{Var}(p_Z). \tag{9}$$

For a window of size m with exchangeable correctness indicators, marginal correctness probability $\bar{p}$, and pairwise correlation ρ, summing variances and covariances gives

$$\mathrm{Var}(\bar{Y}) = \frac{\bar{p}(1-\bar{p})}{m}[1 + (m-1)\rho].$$

For $0 < \bar{p} < 1$ and a positive variance inflation factor, this corresponds to $m/[1 + (m-1)\rho]$ independent Bernoulli observations with the same variance. This conditional-mixture illustration is not an estimate of the effective sample size of a full path-consistency run. Neither that value nor a plug-in correction is used to calibrate the actual path-consistency gate.

B Additional analysis of prefix reuse

The following results clarify when prefix reuse can affect the final vote and how shared prefixes change the relationship between trajectory accuracy and voting accuracy.

B.1 When can later windows still change the vote?

Let c_a count all valid votes for answer a already generated, including votes from rejected windows. Let ℓ be a unique current leader and let r be an upper bound on the number of answers still to be generated. Include an unseen competitor with count zero when defining $c_{(2)} = \max_{a \neq \ell} c_a$. Each remaining trajectory supplies at most one unit vote, and earlier votes cannot be removed or reweighted.

Proposition 3 (Irreversible plurality)**.** *If $c_\ell - c_{(2)} > r$, then ℓ is the unique final plurality winner for every possible continuation, including adaptive and dependent continuations. If $c_\ell - c_{(2)} < r$, a continuation assigning every remaining vote to a strongest competitor can overturn the leader. At equality a competitor can tie, so the outcome depends on the fixed tie rule.*

Proof. For any competitor, the largest possible increase in its count relative to the leader is r. Thus its final deficit is at least $c_\ell - c_a - r$. A strictly positive minimum guarantees the leader remains unique. Conversely, giving all r votes to a strongest competitor realizes that maximum increase, proving the remaining statements. Invalid outputs can only reduce the number of added votes and cannot break the sufficient condition. □

The statement concerns the realized vote, not correctness. Once a wrong leader is locked, a rollback cannot repair the final answer under this aggregation rule. Once a correct leader is locked, later prefixes cannot change final correctness either. For a fixed $B = 32, m = 4$, one may test the sufficient condition after window j using $r = 32 - 4j$. This is a diagnostic of the final vote, not a stopping rule used in the experiments. The condition says only that later votes cannot alter the outcome; it does not undo any effect of earlier prefixes on the current counts.

There is also a useful consequence for Theorem 1. On a locked history, V^S is constant over all admissible next windows, so $d_j(h) = 0$. Accordingly,

$$|A_P - A_S| \le \min\left\{1, \sum_j \mathbb{E}_P\left[\mathbf{1}\{H_j \text{ is not locked}\}\ \mathrm{TV}(K_j^P(\cdot \mid H_j), K_j^S(\cdot \mid H_j))\right]\right\}.$$

This refinement localizes possible final-vote effects to histories where future votes can still matter. It does not assert that unlocked histories are recoverable under the model's actual support, nor that their increments are favorable.

B.2 Higher marginal correctness can yield a worse majority

For a binary illustration, condition on a history with n earlier votes, k of them correct, and $r = B - n$ remaining votes. A random shared prefix Z induces a correctness probability p_Z; conditional on Z and the history, the remaining answers are independent Bernoulli variables. With ties counted as failures, let $t = \lfloor B/2 \rfloor + 1 - k$. Using the binomial tail from Equation 7, the conditional final success probability is

$$A_{\text{shared}}(h) = \mathbb{E}[Q_r(p_Z, t) \mid h]. \tag{10}$$

If an independent prefix were instead drawn for each remaining answer under this stylized model, the probability would be $Q_r(\bar{p}, t)$, where $\bar{p} = \mathbb{E}[p_Z \mid h]$. These two expressions generally differ even at identical marginal correctness. Independent-prefix resampling here is an analytical comparator, not an evaluated path-consistency variant.

Proposition 4 (Curvature of the voting response)**.** *For $1 \le t \le r$ and $0 < p < 1$,*

$$\frac{dQ_r(p,t)}{dp} = r\binom{r-1}{t-1}p^{t-1}(1-p)^{r-t},$$

and the sign of its second derivative is the sign of $(t-1) - (r-1)p$. Thus Q_r is concave above $p = (t-1)/(r-1)$ and convex below it when $r > 1$. If the support of p_Z lies wholly in the concave region, sharing cannot improve the voting probability relative to independent resampling at the same mean; in the convex region the inequality reverses.

Proof. Differentiate the finite binomial sum. Adjacent terms cancel, leaving the stated first derivative. Its logarithmic derivative is $(t-1)/p - (r-t)/(1-p)$, which has the stated sign. Jensen's inequality on either curvature region gives the comparison. For $r = 1$ the response is linear. For $t \le 0$ or $t > r$ it is constant. When the support crosses the inflection point, these local curvature statements imply no universal ordering. □

For a finite-budget counterexample retaining the root votes, take four independent root answers with correctness probability 0.6 and 28 later answers sharing a latent prefix. Independently of the root, let

$p_Z = 1$ with probability 0.65 and $p_Z = 0$ otherwise. The final vote is correct exactly when $p_Z = 1$, hence its accuracy is 0.65. In contrast, 32 independent answers with correctness probability 0.6 succeed with probability $Q_{32}(0.6, 17) = 0.835203095$. Every later answer in the shared construction has higher marginal correctness (0.65 versus 0.6), and the mean across all 32 is 0.64375, yet its majority is worse. This deliberately extreme mixture is not a fitted path-consistency model or an assertion that these prefixes occur in the experiments. It demonstrates why per-trajectory correctness and agreement alone do not identify the accuracy of a shared-prefix vote.

B.3 Agreement thresholds under a fixed answer distribution

A fixed-distribution calculation helps distinguish evidence for answer concentration from evidence for correctness. Suppose n answers are sampled independently from one distribution with correct-answer probability q. For $1/2 < C < 1$, define

$$k_C(n) = \min\{k \in \{0, \dots, n\} : c(n, k) \geq C\},$$

taking $k_C(n) = n + 1$ when the set is empty. The score in Equation 1 is increasing in k, and $k_C(n) > n/2$. Hence the correct answer passes the gate exactly when its count reaches $k_C(n)$, with probability

$$\mathbb{P}(\text{correct answer passes}) = Q_n(q, k_C(n)).$$

This expression concerns the source answer, not the semantic correctness of every token in its prefix.

An incorrect answer can pass only if the total number of incorrect answers is at least $k_C(n)$. Therefore

$$\mathbb{P}(\text{an incorrect answer passes}) \leq Q_n(1 - q, k_C(n)).$$

Equality holds when all incorrect probability mass is concentrated on one alternative. This identifies a worst-case error distribution for a single fixed-distribution check. Raising C increases the required count and reduces both correct and incorrect gate-passing probabilities at that check. In particular, a majority of correct answers can still fail a stringent gate, so the probability of failing to pass cannot be bounded solely by the probability of lacking a correct majority.

Successive checks in path-consistency reuse observations and change the sampling distribution. Their error probabilities therefore cannot be multiplied as if the checks were independent. More generally, if the conditional probability of an incorrect gate-passing decision at check j, given its preceding history, is bounded by ε_j, the probability of any such decision is at most $\sum_j \varepsilon_j$ by a union bound and conditional expectation. The gate itself does not establish these conditional bounds. The adaptive analysis in Section 4 avoids treating repeated agreement as a correctness guarantee.

C Method and evaluation details

C.1 Window transitions and answer aggregation

Table 7 summarizes the controller in Section 3. The admitted set $\mathcal{V}$ controls prefix proposals and validation, whereas the final vote uses all valid completed answers. Answers from previously accepted windows remain in $\mathcal{V}$ after rollback.

Failed windows provide neither admitted answers nor new prefix candidates. A failed candidate is not permanently excluded from all later source pools. When combined with an early-stopping method, sampling ends at a completed window according to that method's stopping rule, and its aggregation rule is retained.

C.2 Datasets and prompting

We use the GSM8K test split, MATH-500, AIME 2024, AIME 2025, and GPQA-Diamond. GPQA option order is shuffled deterministically with seed 2027.

The mathematical reasoning instruction is *Please reason step by step, and put your final answer within* `\boxed{}`. GPQA presents four labeled options and requests the correct option letter in the same format. Llama uses its chat template. DeepSeek and Qwen use thinking mode, with the decoding settings in Section 5.1. These interfaces differ, so comparisons are made within each model and dataset.

Table 7: Prefix updates at window boundaries. Every valid completed answer remains eligible for final aggregation.

State	Observation	Update
Root	Valid answers available	Add answers to $\mathcal{V}$; propose an extension if the gate passes and a candidate exists.
Trial	Same modal answer and $c \geq C$	Commit the prefix, admit answers, clear the failure count, and consider the next extension.
Trial	Modal answer changes or $c < C$	Discard the proposal, return to the committed prefix, and increment the failure count.
Committed	Same modal answer and $c \geq C$	Retain the prefix, admit answers, clear the failure count, and consider the next extension.
Committed	Modal answer changes or $c < C$	Roll back one level and increment the failure count.
Any prefix	Two consecutive failures	Clear the prefix and disable further reuse for this question.

C.3 Token and time accounting

At an empty prefix, path-consistency reuses the corresponding self-consistency window, including after a return to root. Its original token count and generation time are charged in full. Windows with nonempty prefixes generate new continuations, and failed windows are also counted.

Let T_{ij} and G_{ij} be generated tokens and generation time for window j of question i. Over n questions, the means are

$$\bar{T} = \frac{1}{n}\sum_{i,j} T_{ij}, \qquad \bar{G} = \frac{1}{n}\sum_{i,j} G_{ij}.$$

Token savings are $1 - \bar{T}_{\mathrm{PC}}/\bar{T}_{\mathrm{SC}}$, and a generation speed ratio is $\bar{G}_{\mathrm{SC}}/\bar{G}_{\mathrm{PC}}$. Tables with negative Save entries instead display the signed relative token change. A nominal prefix fraction is not a measured saving because continuation lengths can change. These measurements do not include all input-processing and orchestration costs or establish peak-memory and API-billing savings.

C.4 Ablation settings

The prefix-selection and recovery ablation fixes the 490 DeepSeek/MATH-500 questions, self-consistency root windows, source eligibility, prefix-length grid, and decoding settings. Uniform selection samples from the same eligible candidates. Disabling recovery retains the active prefix after failure and disables the two-failure reset, while failed-window answers remain excluded from $\mathcal{V}$. All valid answers still enter the final vote. This removes the recovery mechanism as a whole rather than isolating a single rollback operation.

The full configuration and the three alternatives in Table 5 have 428, 426, 423, and 423 correct answers. Relative to the full configuration, the alternatives have paired gains/losses of 3/5, 4/9, and 5/10. These single-run differences describe the observed contribution of each component rather than establishing its variability across repeated runs.

Agreement threshold. Table 6 compares $C \in \{0, 0.7, 0.8, 1\}$ on DeepSeek/GSM8K. At $C = 0$, the gate imposes no minimum agreement score, but a valid modal answer and feasible prefix are still required. At $C = 1$, prefix reuse is disabled. Lower values favor token savings, while higher values make reuse more conservative and favor accuracy preservation. The default remains $C = 0.8$; this sensitivity analysis does not identify a universally optimal threshold.

Prefix length at a fixed history. A supplementary DeepSeek/MATH-500 study fixes the first-window history and the source chosen by local confidence selection, then varies the retained fraction over $\{0, 0.2, 0.4, 0.6, 0.8\}$. It covers 466 questions with feasible prefixes at every level and eight repetitions of a four-answer window per level. Of the remaining questions, 19 do not pass the initial gate and five lack a feasible endpoint. The final-vote diagnostic combines the intervened window with the same four root answers and 24 unprefixed answers. It therefore isolates one window under

a fixed continuation, rather than estimating the full continuation expectation in Equation 4. This analysis does not change the fixed prefix-length grid used at inference time.

D INITIAL EXPLORATION OF SENTENCE-LEVEL PREFIX REUSE

Before developing the current method, we first explored a sentence-level form of path-consistency for short-chain reasoning. The idea was to reuse complete early reasoning steps rather than generate every trajectory from scratch. This initial approach achieved substantial token savings with generally comparable accuracy on several arithmetic, commonsense, and symbolic tasks, motivating the controlled prefix reuse developed in Section 3. We summarize these earlier experiments and the insights they provided below.

The sentence-level approach samples 20 trajectories and randomly selects from eligible prefix candidates ending at sentence boundaries. Its default schedule allows three extension levels, with updates after every five trajectories. Unlike the current method, it does not use one shared prefix per window, the token-fraction grid, or the same trial-and-recovery controller. The results therefore characterize this initial approach and are not pooled with the main experiments.

The Llama-3-8B experiments use temperature 0.6, top-$p = 0.9$, few-shot reasoning prompts, and an RTX 3090. Here a prefix level counts retained reasoning steps rather than a fraction of trajectory tokens. Comparisons retain their own self-consistency baselines and timing conventions. Reported speedup percentages denote the increase in speed relative to the corresponding baseline, not the percentage reduction in elapsed time.

D.1 ARITHMETIC, COMMONSENSE, AND SYMBOLIC REASONING

Table 8 reports our initial results across ten short-chain reasoning tasks (Cobbe et al., 2021; Patel et al., 2021; Miao et al., 2020; Roy & Roth, 2015; Geva et al., 2021; Suzgun et al., 2023). Sentence-level reuse reduces token generation across all tasks and thresholds, with savings ranging from 4.7% to 36.6%. Arithmetic accuracy is generally close to or above the self-consistency baseline. At $C = 0.8$, GSM8K and Tracking Shuffled Objects improve by 3.7 and 5.2 percentage points, respectively, while Ruin Names and Logical Deduction decline. These results established the practical potential of prefix reuse while showing that its accuracy effects depend on the task.

Table 8: Sentence-level prefix reuse with Llama-3-8B. Each PC entry gives accuracy (%) / token saving (%) relative to its SC baseline. These results use the supplementary 20-trajectory setting.

Dataset	SC accuracy (%)	$C = 0$	$C = 0.7$	$C = 0.8$	$C = 0.9$
Arithmetic reasoning					
GSM8K	64.1	66.6 / 24.0	67.1 / 16.1	67.8 / 16.2	66.6 / 9.9
SVAMP	79.3	78.8 / 36.6	78.5 / 29.4	79.2 / 29.7	79.9 / 22.3
ASDiv	81.0	80.1 / 32.7	81.2 / 27.2	80.0 / 27.1	80.6 / 22.6
MultiArith	96.7	97.8 / 34.8	97.2 / 33.3	98.9 / 33.1	98.3 / 27.2
Commonsense reasoning					
StrategyQA	71.4	70.8 / 27.8	70.6 / 24.0	71.3 / 22.1	71.1 / 16.0
Ruin Names	76.0	71.2 / 9.8	74.4 / 7.2	72.0 / 6.6	71.6 / 4.7
Salient Translation	54.8	58.4 / 25.8	57.2 / 18.3	56.4 / 16.4	55.2 / 10.6
Symbolic reasoning					
Boolean Expressions	88.4	90.0 / 21.3	89.6 / 19.8	88.8 / 20.3	89.6 / 18.3
Tracking Shuffled Objects	54.0	58.0 / 16.7	56.0 / 12.4	59.2 / 10.8	58.4 / 6.8
Logical Deduction	78.8	76.8 / 26.1	78.8 / 20.5	77.2 / 24.8	76.0 / 17.5

Lower thresholds generally permit larger savings, but the trend need not be monotonic after the generated history changes. Logical Deduction, for example, saves more tokens at $C = 0.8$ than at $C = 0.7$. The smaller improvements on Ruin Names also show that the benefit depends on the task, rather than only on the trajectory budget.

Table 9: Reported generation speedups (%) for the sentence-level experiments in Table 8, relative to each task's SC baseline.

Dataset	$C = 0$	$C = 0.7$	$C = 0.8$	$C = 0.9$
GSM8K	28.9	18.0	17.4	10.2
SVAMP	48.3	36.6	35.7	24.8
ASDiv	42.0	32.1	31.1	24.9
MultiArith	42.7	40.8	40.5	33.8
StrategyQA	34.2	29.4	24.3	16.4
Ruin Names	10.4	7.8	7.4	4.7
Salient Translation	30.5	19.6	17.3	10.2
Boolean Expressions	26.6	24.3	25.2	22.0
Tracking Shuffled Objects	18.4	12.9	11.3	6.8
Logical Deduction	31.6	23.1	30.1	19.1

D.2 Prefix growth and extension frequency

Table 10 shows how savings grow with sentence-level prefixes on GSM8K. At $C = 0.8$, token savings rise from 9.8% at level 1 to 33.3% at level 3. This observation motivated progressive prefix extension in the current method, where prefix length is expressed as a fraction of reasoning tokens rather than a number of sentences.

Table 10: GSM8K efficiency by sentence-prefix level. The left group reports speedup (%); the right group reports token saving (%), each relative to the corresponding unprefixed baseline.

	Speedup (%)				Save (%)			
C	Overall	Level 1	Level 2	Level 3	Overall	Level 1	Level 2	Level 3
0	28.9	18.8	42.6	78.9	24.0	16.9	32.7	47.4
0.7	18.0	10.0	25.3	46.5	16.1	9.4	21.3	33.8
0.8	17.4	9.8	25.0	43.9	16.2	9.8	21.8	33.3
0.9	10.2	4.2	13.3	27.0	9.9	4.2	12.7	23.0

With the same 20-trajectory budget, allowing four rather than three extension levels also changes the update interval from five trajectories to four. Table 11 shows increased savings under the more frequent schedule. Accuracy is unchanged at $C = 0.8$ and lower at $C = 0.7$. Because depth and update interval change together, this comparison does not isolate the effect of window size.

Table 11: Extension depth and update interval on Llama-3-8B/GSM8K with sentence-level prefixes. Speedup and Save are relative to SC.

C	Max. level	Window size	Accuracy (%)	Speedup (%)	Save (%)
0.7	3	5	67.1	18.0	16.1
0.7	4	4	66.3	20.5	20.0
0.8	3	5	67.8	17.4	16.2
0.8	4	4	67.8	18.0	17.8

D.3 Method comparisons across model families

Table 12 compares sentence-level prefix reuse with early-stopping methods. On Llama-3-8B, path-consistency uses fewer tokens than either early-stopping baseline on all three tasks, with accuracy between or close to the comparison methods. The DeepSeek-V3 and Llama-3.2-1B-Instruct results show the same cost advantage, but the smaller model loses some accuracy on GSM8K and StrategyQA. Together, these experiments suggested that sentence-level reuse was useful across model families, while its accuracy remained constrained by the quality of the underlying reasoning.

Table 12: Supplementary method comparisons. Entries are accuracy (%) / mean tokens per question. Each model block is a separate evaluation. AC and ESC denote adaptive-consistency and early-stopping self-consistency. The Llama-3-8B baseline here differs from that in Table 8.

Model	Method	GSM8K	SVAMP	StrategyQA
Llama-3-8B (Grattafiori et al., 2024)	SC	68.4 / 3,860	80.4 / 2,321	71.7 / 1,805
	AC	67.2 / 2,248	79.7 / 935	71.2 / 706
	ESC	67.2 / 2,710	79.2 / 1,154	70.8 / 782
	PC	67.8 / 1,630	79.2 / 816	71.3 / 700
DeepSeek-V3 (DeepSeek-AI et al., 2024)	AC	91.6 / 828	94.7 / 363	84.5 / 326
	ESC	91.6 / 868	94.7 / 358	84.2 / 315
	PC	91.8 / 786	94.7 / 263	84.5 / 230
Llama-3.2-1B-Instruct	AC	48.1 / 2,748	61.1 / 800	63.9 / 1,401
	ESC	47.9 / 2,324	61.9 / 924	63.7 / 1,136
	PC	47.3 / 1,430	62.0 / 711	63.1 / 799

Table 13 provides another threshold comparison on Mistral-7B. Removing the agreement barrier gives the largest token saving but lower accuracy, while higher thresholds retain accuracy closer to the self-consistency baseline with smaller savings.

Table 13: Sentence-level prefix reuse on Mistral-7B/GSM8K. Speedup and Save are relative to SC.

Method	C	Accuracy (%)	Speedup (%)	Save (%)
SC	1.0	59.7	0.0	0.0
PC	0.0	58.5	22.0	20.3
PC	0.8	60.3	11.1	11.4
PC	0.9	60.4	6.2	6.6

D.4 Token allocation and answer agreement

Our initial token analysis in Figure 4 compared generation on trajectories with correct and incorrect final answers. Sentence-level reuse reduces generated text in both groups, supporting the idea that savings come from shortening later continuations rather than only avoiding incorrect answers. Final-answer labels do not identify which individual reasoning tokens are unnecessary. Figure 1 complements this analysis by directly visualizing repeated token sequences.

We also examined whether answer agreement could guide prefix reuse. Figure 5 shows that stronger agreement after ten self-consistency trajectories is associated with correct answers across several tasks. Figure 6 compares the final distributions on GSM8K, where sentence-level reuse produces more concentrated agreement. Incorrect answers nevertheless occur at high agreement. These observations motivated agreement-guided control, but not the interpretation of agreement as a correctness certificate.

D.5 Illustrative reasoning trajectories

Table 14 illustrates progressive sentence-prefix reuse. Inherited text is retained while the newly generated suffix becomes shorter. It shows how the initial approach reused complete reasoning steps, complementing the token-fraction example of the current method in Table 2.

A second example asks how much water remains after two girls each receive one sixth of 24 liters and a boy receives 6 liters. The correct answer is 10 liters. Table 15 summarizes the Llama-3.2-1B-Instruct trajectories. Prefix levels increase and later return to zero; the final vote is correct despite earlier disagreement. The example shows that a return to unprefixed sampling can occur, but does not prove that the reused text was wrong or that recovery caused the correct final answer.

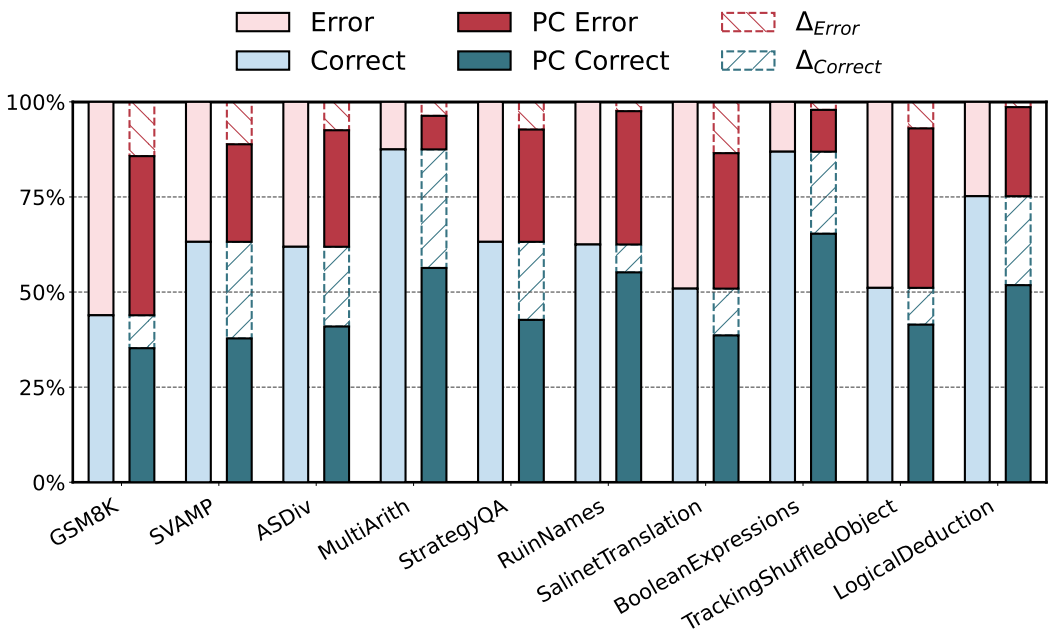


Figure 4: Token allocation in the initial sentence-level experiments. Light bars show SC, dark bars show PC, and hatched areas indicate reductions relative to SC. Correct and Error classify trajectories by their final answers, not individual tokens.

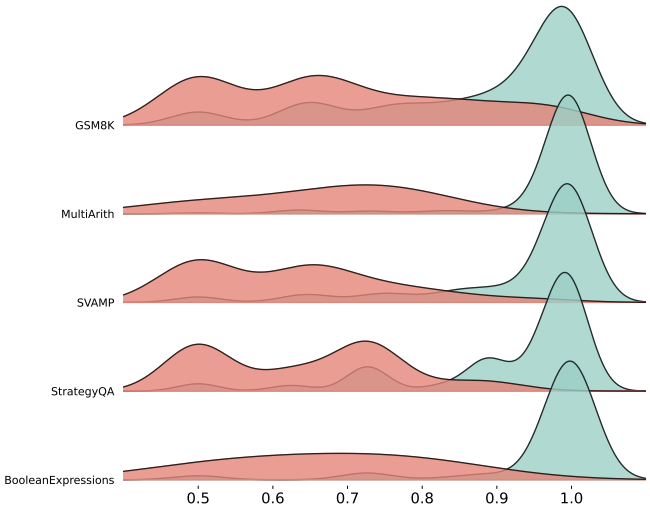
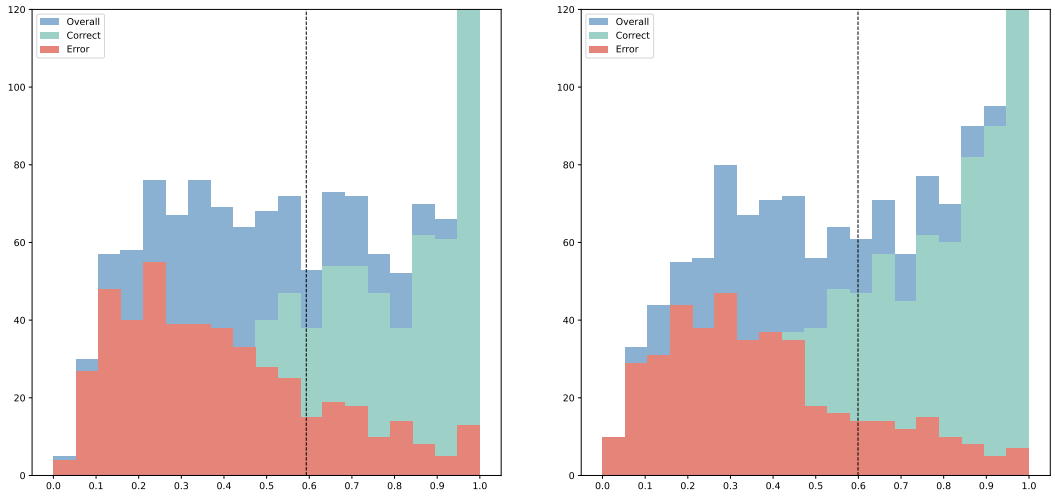


Figure 5: Answer agreement after ten SC trajectories in the initial experiments. Green and red show the distributions for correct and incorrect answers, respectively.

D.6 Controller overhead and parallelism

The sentence-level Llama-3-8B/GSM8K measurements in Table 16 show that agreement evaluation and prefix selection take little time relative to model generation in that setting. They do not measure the full controller cost of the current method, which also ranks local confidence scores and manages trial and recovery.

Windowed reuse also constrains parallelism. Trajectories within a window can run concurrently, but the next window depends on the preceding answers and prefix decision. Larger windows allow



(a) Self-consistency

(b) Path-consistency

Figure 6: Final answer agreement on GSM8K in the sentence-level experiments, with $C = 0.8$ for PC. Blue shows all questions, green correct answers, and red incorrect answers. The panels use the same axis scales.

Table 14: A sentence-level prefix example. Black text is reused and blue text is newly generated.

Question	A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?
No prefix	To make a robe, we need 2 bolts of blue fiber and half that much white fiber. We can write this as 2 + 1/2 * 2. Now 1/2 * 2 = 1, so the equation becomes 2 + 1 = 3. The answer is 3.
Level 1	To make a robe, we need 2 bolts of blue fiber and half that much white fiber. 2 * 1/2 = 1 bolt of white fiber. 2 + 1 = 3 bolts in total. The answer is 3.
Level 2	To make a robe, we need 2 bolts of blue fiber and half that much white fiber. 2 * 1/2 = 1 bolt of white fiber. 1 + 2 = 3 bolts in total. The answer is 3.
Level 3	To make a robe, we need 2 bolts of blue fiber and half that much white fiber. 2 * 1/2 = 1 bolt of white fiber. 1 + 2 = 3 bolts in total. The answer is 3.

Table 15: Window-level summary of the water-allocation example. Tokens are the total newly generated tokens in each five-trajectory window. The recorded final answer is 10; answers 10 and 14 each receive seven votes.

Window	Prefix level	Sampled answers	Tokens
1	0	8, 10, 14, 14, 14	383
2	1	2, 14, 14, 10, 16	283
3	2	10, 10, 10, 10, 12	89
4	0	16, 14, 2, 10, 14	380

Table 16: Controller timing in the supplementary 20-trajectory Llama-3-8B/GSM8K evaluation. Percentages are relative to the reported model-generation time.

Component	Time (s)	Relative time (%)
Model generation	2.2	100
Answer agreement evaluation	0.000096	0.0044
Prefix construction	0.00013	0.0059

more concurrent generation, whereas smaller windows offer more frequent opportunities to extend a prefix. Token savings therefore need not translate into proportional latency reductions, especially when independent self-consistency trajectories can be served with greater parallelism.

D.7 Prompt structure and scope

The sentence-level experiments use few-shot demonstrations containing a question, reasoning, and final answer. Table 17 summarizes their format without repeating the full demonstration sets. These prompts are distinct from the chat and thinking interfaces used in the main evaluation.

Table 17: Prompt structure for the supplementary sentence-level experiments. Demonstrations supply intermediate reasoning followed by an explicit final answer.

Task family	Demonstration content	Answer format
Arithmetic reasoning	Arithmetic steps leading to a numerical result	The answer is [number].
StrategyQA	Short factual reasoning for a yes/no question	The answer is yes/no.
Ruin Names	Comparison of candidate humorous edits	The answer is [option].
Salient Translation	Comparison of a source sentence and its translation	The answer is [error-type option].
Boolean Expressions	Evaluation using logical operator precedence	The answer is True/False.
Tracking Shuffled Objects	State updates after successive swaps	The answer is [option].
Logical Deduction	Ordering objects from the supplied constraints	The answer is [option].

Implications for the current method. The initial experiments showed that reusing intermediate reasoning can shorten later trajectories across several short-chain tasks. They also exposed the trade-off between more aggressive reuse and accuracy preservation. The current method builds on this idea with one continuous shared prefix per window, a token-fraction length policy, local confidence selection, and explicit trial and recovery. The earlier results motivate this design direction; the experiments in Section 5 evaluate the resulting method.

These experiments do not train or modify the evaluated models. Their computational benefits and failure modes depend on the available reusable reasoning, the agreement threshold, and the generation interface. Reducing inference cost may improve accessibility but can also reduce the cost of harmful uses of the underlying models. Dataset text and model weights remain subject to their providers' terms and are not redistributed with the supplementary results.